

Experiment-1

Aim: Study of different types of Network cables and Practically implement the cross-wired cable and straight through cable using clamping tool.

Apparatus (Components): RJ-45 connector, Clipping Tool, Twisted pair Cable

Procedure: To do these practical following steps should be done:

1. Start by stripping off about 2 inches of the plastic jacket off the end of the cable. Be very careful at this point, as to not nick or cut into the wires, which are inside. Doing so could alter the characteristics of your cable, or even worse render it useless. Check the wires, one more time for nicks or cuts. If there are any, just whack the whole end off, and start over.
2. Spread the wires apart, but be sure to hold onto the base of the jacket with your other hand. You do not want the wires to become untwisted down inside the jacket. Category 5 cable must only have 1/2 of an inch of 'untwisted' wire at the end; otherwise it will be 'out of spec'. At this point, you obviously have ALOT more than 1/2 of an inch of un-twisted wire.
3. You have 2 end jacks, which must be installed on your cable. If you are using a pre-made cable, with one of the ends whacked off, you only have one end to install - the crossed over end. Below are two diagrams, which show how you need to arrange the cables for each type of cable end. Decide at this point which end you are making and examine the associated picture below.

Diagram shows you how to prepare Cross wired connection

RJ45 Pin # (END 1)	Wire Color	Diagram End #1	RJ45 Pin # (END 2)	Wire Color	Diagram End #2
1	White/Orange		1	White/Green	
2	Orange		2	Green	
3	White/Green		3	White/Orange	
4	Blue		4	White/Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/Brown		7	Blue	
8	Brown		8	White/Blue	

Diagram shows you how to prepare straight through wired connection

RJ45 Pin # (END 1)	Wire Color	Diagram End #1	RJ45 Pin # (END 2)	Wire Color	Diagram End #2
1	White/Orange		1	White/Green	
2	Orange		2	Green	
3	White/Green		3	White/Orange	
4	Blue		4	White/Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/Brown		7	Blue	
8	Brown		8	White/Blue	

Experiment - 2

Aim: Study of following Network Devices in Detail

- Repeater
- Hub
- Switch
- Bridge
- Router
- Gate Way

Apparatus (Software): No software or hardware needed.

Procedure: Following should be done to understand this practical.

1. **Repeater:** Functioning at Physical Layer. A **repeater** is an electronic device that receives a signal and retransmits it at a higher level and/or higher power, or onto the other side of an obstruction, so that the signal can cover longer distances. Repeater have two ports ,so cannot be use to connect for more than two devices

2. **Hub:** An **Ethernet hub, active hub, network hub, repeater hub, hub** or **concentrator** is a device for connecting multiple twisted pair or fiber optic Ethernet devices together and making them act as a single network segment. Hubs work at the physical layer (layer 1) of the OSI model. The device is a form of multiport repeater. Repeater hubs also participate in collision detection, forwarding a jam signal to all ports if it detects a collision.

3. **Switch:** A **network switch** or **switching hub** is a computer networking device that connects network segments. The term commonly refers to a network bridge that processes and routes data at the data link layer (layer 2) of the OSI model. Switches that additionally process data at the network layer (layer 3 and above) are often referred to as Layer 3 switches or multilayer switches.

4. **Bridge:** A **network bridge** connects multiple network segments at the data link layer (Layer 2) of the OSI model. In Ethernet networks, the term *bridge* formally means a device that behaves according to the IEEE 802.1D standard. A bridge and switch are very much alike; a switch being a bridge with numerous ports. *Switch* or *Layer 2 switch* is often used interchangeably with *bridge* .Bridges can analyze incoming data packets to determine if the bridge is able to send the given packet to another segment of the network.

5. **Router:** A **router** is an electronic device that interconnects two or more computer networks, and selectively interchanges packets of data between them. Each data packet contains address information that a router can use to determine if the source and destination are on the same network, or if the data packet must be transferred from one network to another. Where multiple routers are used in a large collection of interconnected networks, the routers exchange information about target system addresses, so that each router can build up a table showing the preferred paths between any two systems on the interconnected networks.

6. **Gate Way:** In a communications network, a network node equipped for interfacing with

another network that uses different protocols.

- A gateway may contain devices such as protocol translators, impedance matching devices, rate converters, fault isolators, or signal translators as necessary to provide system interoperability. It also requires the establishment of mutually acceptable administrative procedures between both networks.
- A protocol translation/mapping gateway interconnects networks with different network protocol technologies by performing the required protocol conversions.

Experiment - 3

Aim: Study of network IP

- Classification of IP address
- Sub netting
- Super netting

Apparatus (Software): NA

Procedure: Following is required to be study under this practical.

- Classification of IP address

As show in figure we teach how the ip addresses are classified and when they are used.

Class	Address Range	Supports
Class A	1.0.0.1 to 126.255.255.254	Supports 16 million hosts on each of 127 networks.
Class B	128.1.0.1 to 191.255.255.254	Supports 65,000 hosts on each of 16,000 networks.
Class C	192.0.1.1 to 223.255.254.254	Supports 254 hosts on each of 2 million networks.
Class D	224.0.0.0 to 239.255.255.255	Reserved for multicast groups.
Class E	240.0.0.0 to 254.255.255.254	Reserved.

• Sub netting

Why we Develop sub netting and How to calculate subnet mask and how to identify subnet address.

• Super netting

Why we develop super netting and How to calculate supernet mask and how to identify supernet address.

Experiment-4

Aim: Connect the computers in Local Area Network.

Procedure: On the host computer

On the host computer, follow these steps to share the Internet connection:

1. Log on to the host computer as Administrator or as Owner.
2. Click **Start**, and then click **Control Panel**.
3. Click **Network and Internet Connections**.
4. Click **Network Connections**.
5. Right-click the connection that you use to connect to the Internet. For example, if you connect to the Internet by using a modem, right-click the connection that you want under Dial-up / other network available.
6. Click **Properties**.
7. Click the **Advanced** tab.
8. Under **Internet Connection Sharing**, select the **Allow other network users to connect through this computer's Internet connection** check box.
9. If you are sharing a dial-up Internet connection, select the **Establish a dial-up connection whenever a computer on my network attempts to access the Internet** check box if you want to permit your computer to automatically connect to the Internet.
10. Click **OK**. You receive the following message:

When Internet Connection Sharing is enabled, your LAN adapter will be set to use IP address 192.168.0. 1. Your computer may lose connectivity with other computers on your network. If these other computers have static IP addresses, it is a good idea to set them to obtain their IP addresses automatically. Are you sure you want to enable Internet Connection Sharing?

11. Click **Yes**.

The connection to the Internet is shared to other computers on the local area network (LAN).

The network adapter that is connected to the LAN is configured with a static IP address of 192.168.0. 1 and a subnet mask of 255.255.255.0

On the client computer

To connect to the Internet by using the shared connection, you must confirm the LAN adapter IP configuration, and then configure the client computer. To confirm the LAN adapter IP configuration, follow these steps:

1. Log on to the client computer as Administrator or as Owner.
2. Click **Start**, and then click **Control Panel**.

3. Click **Network and Internet Connections**.
4. Click **Network Connections**.
5. Right-click **Local Area Connection** and then click **Properties**.
6. Click the **General** tab, click **Internet Protocol (TCP/IP)** in the **connection uses the following items** list, and then click **Properties**.

7. In the **Internet Protocol (TCP/IP) Properties** dialog box, click **Obtain an IP address automatically** (if it is not already selected), and then click **OK**.

Note: You can also assign a unique static IP address in the range of 192.168.0.2 to 254. For example, you can assign the following static IP address, subnet mask, and default gateway:

8. IP Address 192.168.31.202
9. Subnet mask 255.255.255.0
10. Default gateway 192.168.31.1

11. In the **Local Area Connection Properties** dialog box, click **OK**.

12. Quit Control Panel.

Experiment-5

Aim: Study of basic network command and Network configuration commands.

Apparatus (Software): Command Prompt And Packet Tracer.

Procedure: To do this EXPERIMENT- follows these steps:

In this EXPERIMENT- students have to understand basic networking commands e.g ping, tracert etc.

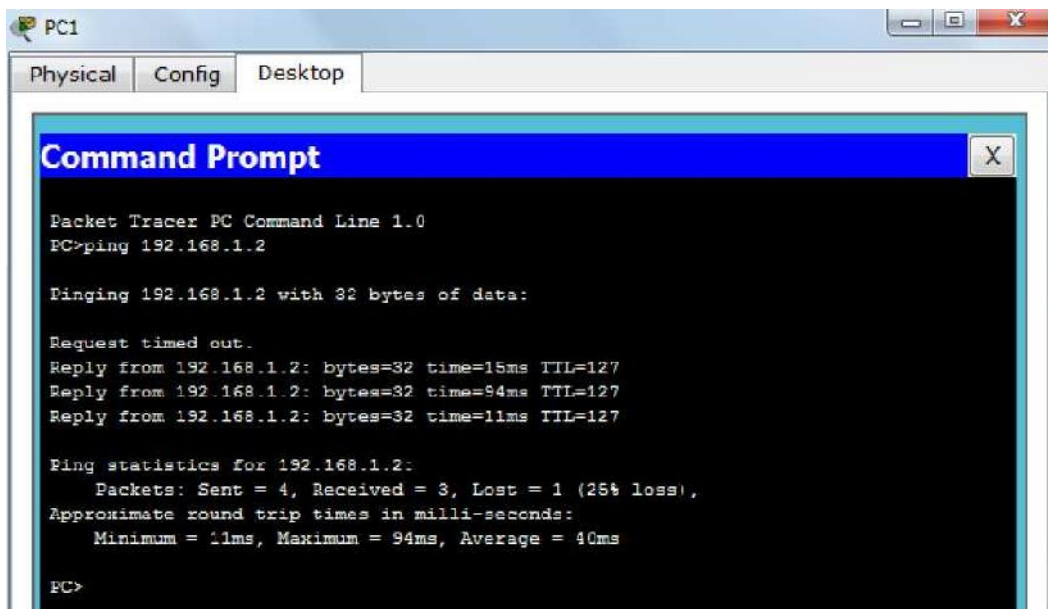
All commands related to Network configuration which includes how to switch to privilege mode and normal mode and how to configure router interface and how to save this configuration to flash memory or permanent memory.

This commands includes

- Configuring the Router commands
- General Commands to configure network
- Privileged Mode commands of a router
- Router Processes & Statistics
- IP Commands
- Other IP Commands e.g. show ip route etc.

ping:

ping(8) sends an ICMP ECHO_REQUEST packet to the specified host. If the host responds, you get an ICMP packet back. Sound strange? Well, you can “ping” an IP address to see if a machine is alive. If there is no response, you know something is wrong.



```
PC1
Physical Config Desktop
Command Prompt
Packet Tracer PC Command Line 1.0
PC>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

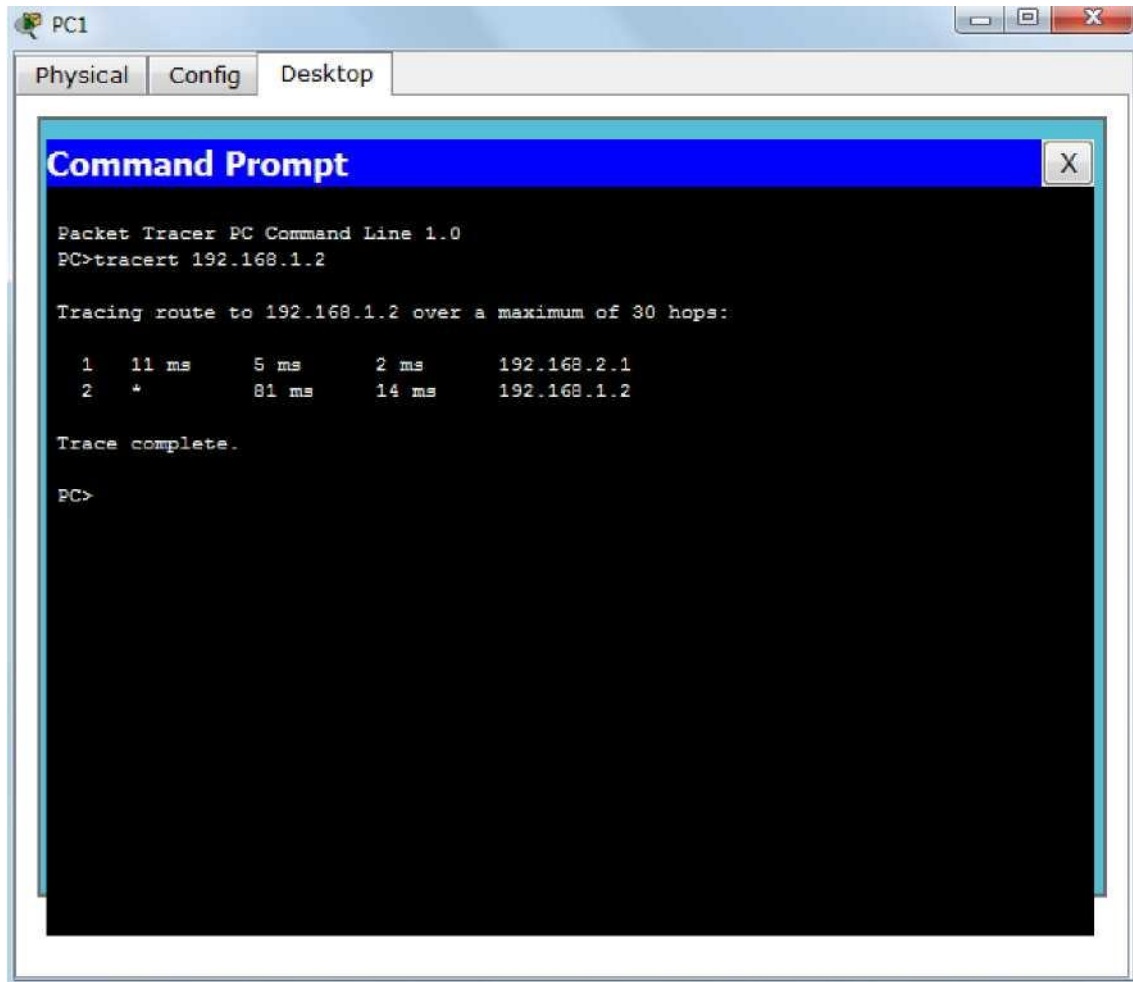
Request timed out.
Reply from 192.168.1.2: bytes=32 time=15ms TTL=127
Reply from 192.168.1.2: bytes=32 time=94ms TTL=127
Reply from 192.168.1.2: bytes=32 time=11ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 94ms, Average = 40ms

PC>
```


Traceroute:

Tracert is a command which can show you the path a packet of information takes from your computer to one you specify. It will list all the routers it passes through until it reaches its destination, or fails to and is discarded. In addition to this, it will tell you how long each 'hop' from router to router takes.



The screenshot shows a Packet Tracer PC window titled 'PC1' with tabs for 'Physical', 'Config', and 'Desktop'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the execution of the 'tracert 192.168.1.2' command, which successfully traces the route to the destination IP address.

```
Packet Tracer PC Command Line 1.0
PC>tracert 192.168.1.2

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.2.1
  1  11 ms   5 ms    2 ms    192.168.2.1
  2  *       81 ms   14 ms   192.168.1.2

Trace complete.

PC>
```

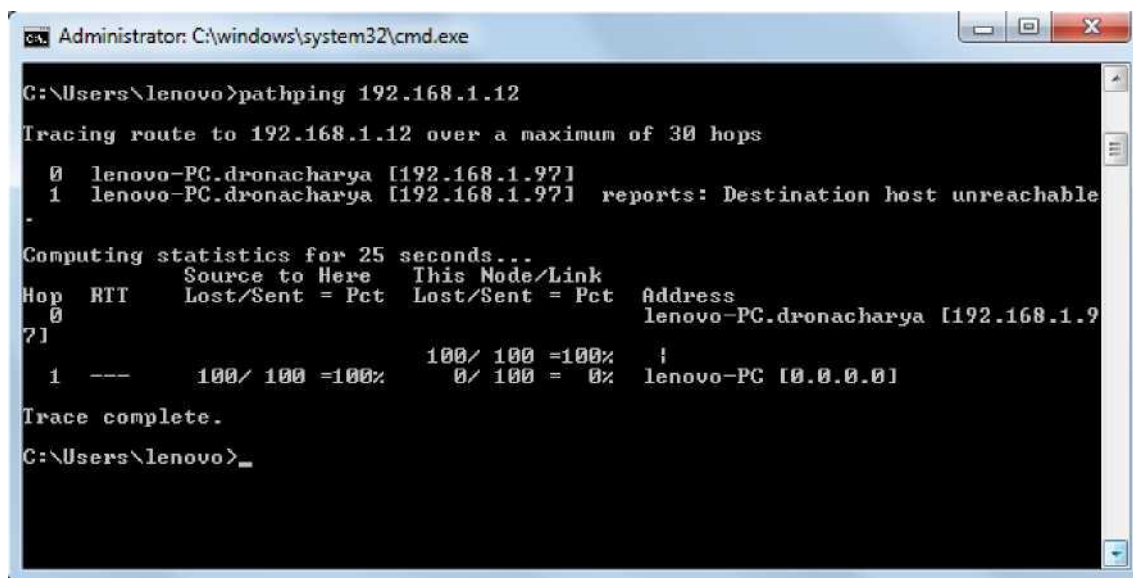
nslookup:

Displays information from Domain Name System (DNS) name servers.

NOTE :If you write the command as above it shows as default your pc's server name firstly.

pathping:

A better version of tracer that gives you statistics about packet lost and latency.



```
Administrator: C:\windows\system32\cmd.exe

C:\Users\lenovo>pathping 192.168.1.12

Tracing route to 192.168.1.12 over a maximum of 30 hops

  0  lenovo-PC.dronacharya [192.168.1.97]
  1  lenovo-PC.dronacharya [192.168.1.97]  reports: Destination host unreachable
-

Computing statistics for 25 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
0
7]
1  ---      100/ 100 =100%    0/ 100 = 0%      lenovo-PC [0.0.0.0]

Trace complete.
C:\Users\lenovo>
```

Getting Help

In any command mode, you can get a list of available commands by entering a question mark (?).

Router>?

To obtain a list of commands that begin with a particular character sequence, type in those characters followed immediately by the question mark (?).

Router#co?

configure connect copy

To list keywords or arguments, enter a question mark in place of a keyword or argument. Include a space before the question mark.

Router#**configure ?**

memory Configure from NV memory network Configure from a TFTP network host terminal
Configure from the terminal

You can also abbreviate commands and keywords by entering just enough characters to make the command unique from other commands. For example, you can abbreviate the **show** command to **sh**.

Configuration Files

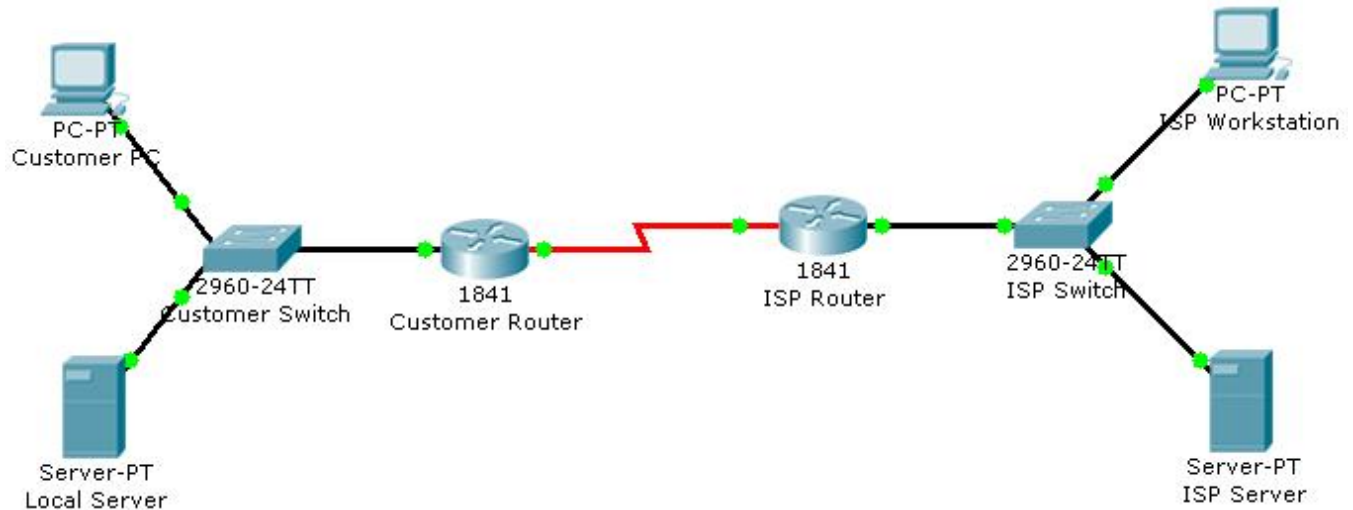
Any time you make changes to the router configuration, you must save the changes to memory because if you do not they will be lost if there is a system reload or power outage. There are two types of configuration files: the running (current operating) configuration and the startup configuration.

Use the following privileged mode commands to work with configuration files.

Experiment-6

Performing an Initial Switch Configuration

Topology Diagram



Objectives

- Perform an initial configuration of a Cisco Catalyst 2960 switch.

Background / Preparation

In this activity, you will configure these settings on the customer Cisco Catalyst 2960 switch:

- Host name
- Console password
- vty password
- Privileged EXEC mode password
- Privileged EXEC mode secret
- IP address on VLAN1 interface
- Default gateway

Note: Not all commands are graded by Packet Tracer.

Step 1: Configure the switch host name.

- From the Customer PC, use a console cable and terminal emulation software to connect to the console of the customer Cisco Catalyst 2960 switch.
- Set the host name on the switch to **CustomerSwitch** using these commands.

```
Switch>enable
Switch#configure terminal
```

```
Switch(config)#hostname CustomerSwitch
```

Step 2: Configure the privileged mode password and secret.

- a. From global configuration mode, configure the password as **cisco**.

```
CustomerSwitch(config)#enable password cisco
```

- b. From global configuration mode, configure the secret as **cisco123**.

```
CustomerSwitch(config)#enable secret cisco123
```

Step 3: Configure the console password.

- a. From global configuration mode, switch to configuration mode to configure the console line.

```
CustomerSwitch(config)#line console 0
```

- b. From line configuration mode, set the password to **cisco** and require the password to be entered at login.

```
CustomerSwitch(config-line)#password cisco
CustomerSwitch(config-line)#login
CustomerSwitch(config-line)#exit
```

Step 4: Configure the vty password.

- a. From global configuration mode, switch to the configuration mode for the vty lines 0 through 15.

```
CustomerSwitch(config)#line vty 0 15
```

- b. From line configuration mode, set the password to **cisco** and require the password to be entered at login.

```
CustomerSwitch(config-line)#password cisco
CustomerSwitch(config-line)#login
CustomerSwitch(config-line)#exit
```

Step 5: Configure an IP address on interface VLAN1.

From global configuration mode, switch to interface configuration mode for VLAN1, and assign the IP address 192.168.1.5 with the subnet mask of 255.255.255.0.

```
CustomerSwitch(config)#interface vlan 1
CustomerSwitch(config-if)#ip address 192.168.1.5 255.255.255.0
CustomerSwitch(config-if)#no shutdown
CustomerSwitch(config-if)#exit
```

Step 6: Configure the default gateway.

- a. From global configuration mode, assign the default gateway to 192.168.1.1.

```
CustomerSwitch(config)#ip default-gateway 192.168.1.1
```

- b. Click the **Check Results** button at the bottom of this instruction window to check your work.

Step 7: Verify the configuration.

The Customer Switch should now be able to ping the ISP Server at 209.165.201.10. The first one or two pings may fail while ARP converges.

```
CustomerSwitch(config)#end
CustomerSwitch#ping 209.165.201.10
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 209.165.201.10, timeout is 2 seconds:

..!!!

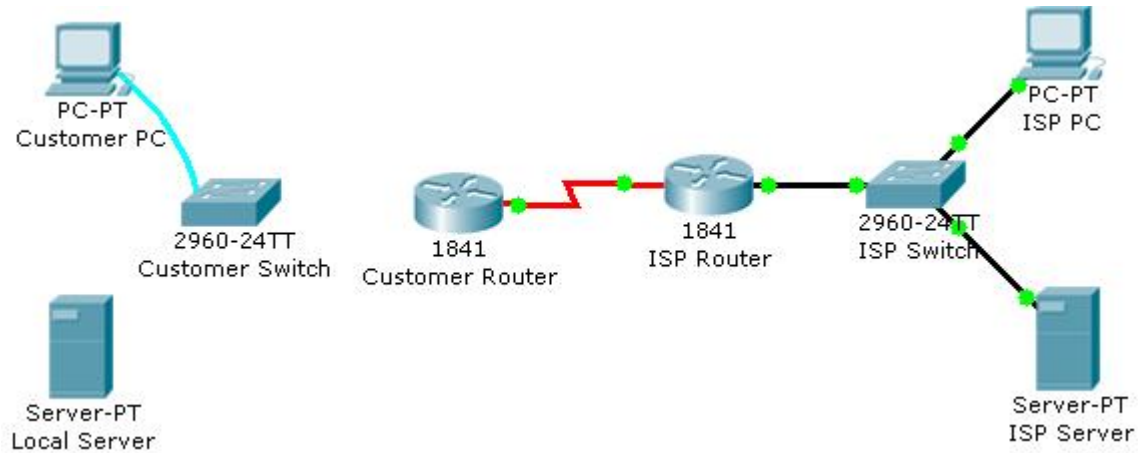
Success rate is 60 percent (3/5), round-trip min/avg/max = 181/189/197 ms

```
CustomerSwitch#
```

Experiment-7

Connecting a Switch

Topology Diagram



Objectives

- Connect a switch to the network.
- Verify the configuration on the switch.

Background / Preparation

In this activity, you will verify the configuration on the customer Cisco Catalyst 2960 switch. The switch is already configured with all the basic necessary information for connecting to the LAN at the customer site. The switch is currently not connected to the network. You will connect the switch to the customer workstation, the customer server, and customer router. You will verify that the switch has been connected and configured successfully by pinging the LAN interface of the customer router.

Step 1: Connect the switch to the LAN.

- Using the proper cable, connect the FastEthernet0/0 on Customer Router to the FastEthernet0/1 on Customer Switch.
- Using the proper cable, connect the Customer PC to the Customer Switch on port FastEthernet0/2.
- Using the proper cable, connect the Local Server to the Customer Switch on port FastEthernet0/3.

Step 2: Verify the switch configuration.

- From the Customer PC, use the terminal emulation software to connect to the console of the customer Cisco Catalyst 2960 switch.
- Use the console connection and terminal utility on the Customer PC to verify the configurations. Use **cisco** as the console password.
- Enter privileged EXEC mode and use the **show running-config** command to verify the following configurations. The password is **cisco123**.
 - VLAN1 IP address = 192.168.1.5
 - Subnet mask = 255.255.255.0

- c. Password required for console access
 - d. Password required for vty access
 - e. Password enabled for privileged EXEC mode
 - f. Secret enabled for privileged EXEC mode
- d. Verify IP connectivity between the Cisco Catalyst 2960 switch and the Cisco 1841 router by initiating a ping to 192.168.1.1 from the switch CLI.
- e. Click the **Check Results** button at the bottom of this instruction window to check your work.

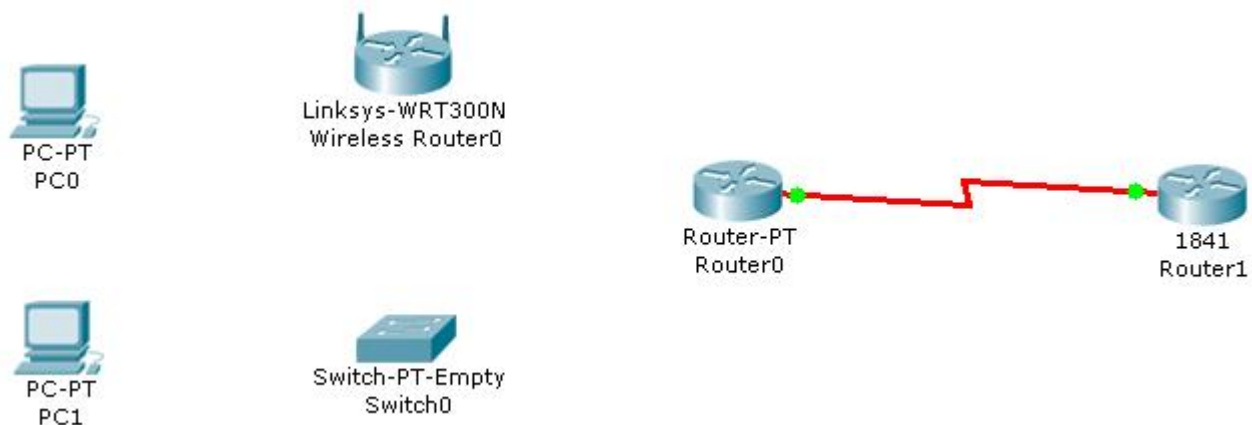
Reflection

- a. What is the significance of the enable secret command compared to the enable password?
- b. If you want to remove the requirement to enter a password to access the console, what commands do you issue from your starting point in privileged EXEC mode?

Experiment- 8

Exploring Different LAN Switch Options

Topology Diagram



Objectives

- Determine the cable types to use to connect all devices to the switch.
- Add appropriate modules to switches and routers.
- Connect the devices to the switch using the appropriate cable types.

Background / Preparation

The results of a site survey for an ISP customer indicate that the customer needs to upgrade the LAN to include a new standalone switch. The network has an existing router (Router0) and a Linksys 300N router. It is necessary to determine which interfaces are needed on the new switch to provide connectivity to the router, the Linksys device, and the customer PCs. The customer wants to use copper cabling.

Note: Links created with the switch may take a minute to change from amber to green. Switch between Simulation mode and Realtime mode to speed up this process.

Step 1: Determine the required connectivity options.

- Click Router0. Using the information in the Physical Device View window on the Physical tab, determine what type of interface is available on the router to connect to the new switch.

Hint: Place the mouse pointer on the interface to display the interface type. Click on the interface type to display a description of the interface.

Which interface is available on the router to connect to the new switch? What type of cable is required?

Click the Linksys 300N. Using the picture on the **Physical** tab, determine what type of cable is necessary to connect to the new switch.

Which interface is available on the Linksys 300N to connect to the new switch? What type of cable is required?

Step 2: Configure the new switch with the required options.

- a. Click Switch0.

On the **Physical** tab, explore each switch module available under the **Modules** option.

Choose the appropriate interfaces to connect to Router0 and the Linksys 300N router.

Choose the appropriate interfaces to connect to the existing PCs.

Power down the switch using the power button in the **Physical Device View** window on the **Physical** tab.

Choose the appropriate modules for the switch. Add the four necessary interfaces to the switch.

Power up the switch using the power button shown in the **Physical Device View** window on the **Physical** tab.

Click the **Config** tab. Select each interface and ensure that the **On** box is checked.

Step 3: Connect the router to the switch.

- a. Using the appropriate cable, connect the router port to the first available switch port. Click the **Config** tab on the router. Select the interface and ensure that the **On** box is checked.
- b. Verify connectivity. A green light appears on each end of the link if the cabling is correct.

Step 4: Connect the Linksys 300N to the switch.

- a. Using the appropriate cable, connect the Linksys 300N to the second available port on the new switch.

Verify connectivity. A green light appears on each end of the link if the cabling is correct.

Step 5: Connect the PCs to the switch.

- a. Using the appropriate cable, connect the existing PCs to the new switch.
- b. Verify connectivity. A green light appears on each end of the links if the cabling is correct.
- c. Click the Check Results button at the bottom of this instruction window to check your work.